

Climate 5000 Series - Gen 4

24K BTU Single Zone Regular System

4-Way Cassette IDU

Engineering
Submittal
Sheet



BOSCH

Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

24K Btu Single Zone Regular System - 4-Way Cassette IDU

Indoor part number	8-733-962-681
Indoor model number	BMS500-AAU024-1AHCXD
Outdoor part number	8-733-962-687
Outdoor model number	BMS500-AAS024-1CSXRD

Efficiency Data		AHRI 210/240 2023
SEER2		22
EER2		12.0
HSPF2 Region IV		10.5
HSPF2 Region V		8.1
COP(47F)		3.20
COP (5F)		2.30

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	12300~26800
	Rated Capacity	24000
	Current	8.8
	Input Power	2000
Heating	Capacity Range	11600~33100
	Rated Capacity (47F)	25000
	Rated Capacity (5F)	18500
	Current	10.2
	Input Power	2268

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 24.0
Max. fuse indoor/outdoor (A)	15.0 / 25.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 620
Input (W)	805 / 2045
Rated current (RLA/A)	4.15 / 9.30

Piping	
Refrigerant precharge (oz/kg)	70.55 / 2.00
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	3/8" / Ø9.52
Gas side (in/mm)	5/8" / Ø15.9
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.32 / 30
Max. refrigerant pipe length (ft/m)	164.00 / 50
Max. difference in height (ft/m)	82.00 / 25

Operation Range

ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		5.41
IDU	Fan motor model	ZKFN-45-8-1
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan speed (RPM)	624 / 568 / 426
	Air flow, Turbo / H / M / L / Si (CFM)	765.2 / 706.3 / 635.7 / 547.4 / N/A
	Approx. Air Throw Range ft (m)	9.8 / 3
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 45.5 / 43.0 / 33.5 / 30.5
	Net weight (lbs / kg)	47.18 / 21.4
	Gross weight (lbs / kg)	59.08 / 26.8
	Panel net weight (lbs / kg)	13.23 / 6
ODU	Gross panel weight (lbs / kg)	19.84 / 9
	Drain pipe diameter (mm)	ODΦ25mm
	Fan motor model	ZKFN-120-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	170.0
	Fan motor speed (RPM)	810 / 700 / 450
	Air flow, Max (CFM)	2236.7
	Noise level, Max (dB(A))	59.5
	Net weight (lbs / kg)	122.13 / 55.4
	Gross weight (lbs / kg)	132.28 / 60.0

Accessories

BMS-RCG-XXD (8733962703)	Decorative Grille - 24K ~ 48K
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK3-7X1 (8733951015)	25ft Lineset

Standard Parts

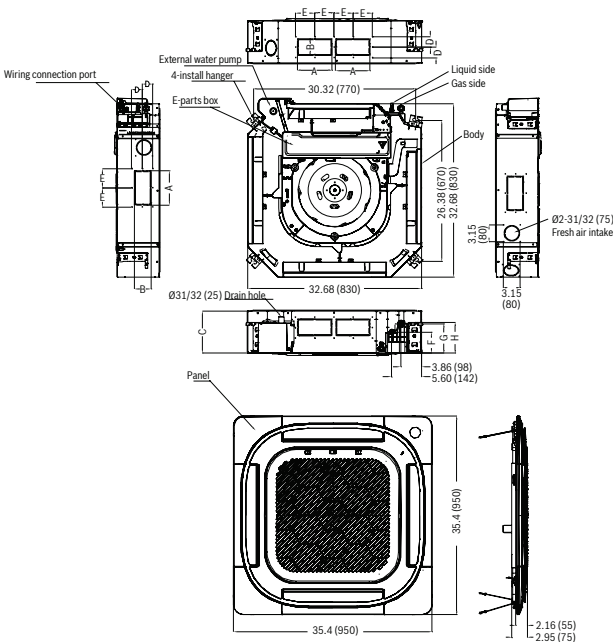
8-733-969-065	Remote Control
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Climate 5000 Series - Gen 4

24K Btu Single Zone Regular System - 4-Way Cassette IDU

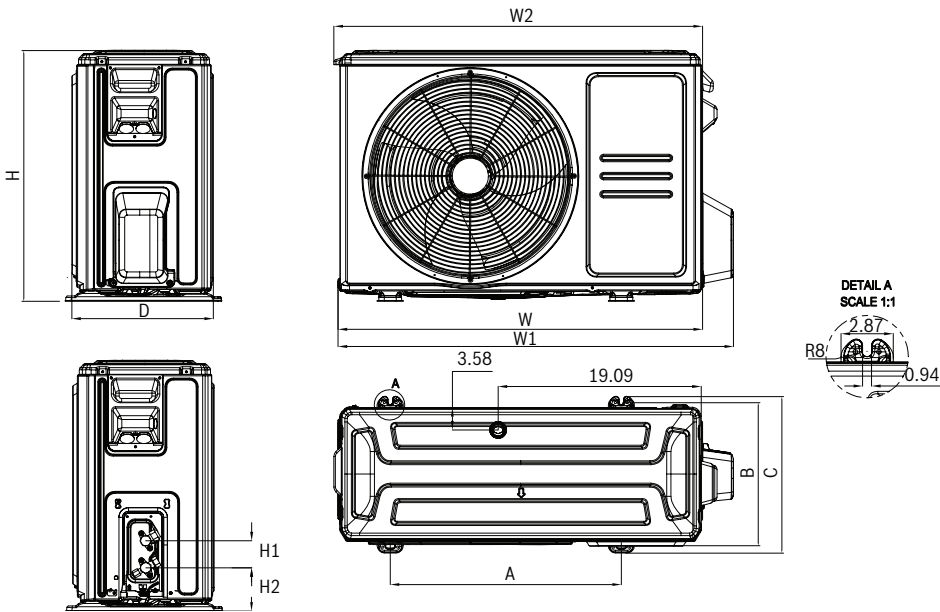


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU024-1AHCXD	32.68	32.68	8.07

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS024-1CSXRD	37.24	16.14	31.89	40.55	37.24	26.5	15.87	17.9	2.36	4.25

Climate 5000 Series - Gen 4

24K Btu Single Zone Regular System - 4-Way Cassette IDU


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Cooling Performance - Indoor Dry Bulb Temperature at 765 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
	S/T	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
	PI	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
-13°F (-25.0°C)	TC	18313	18338	18358	18358	18383	19417	19440	19459	19463	19486	20577	20604	20627	20630	20654	21817	21840	21864	21868	21895
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.76	0.76	0.76	0.76	0.76	0.81	0.81	0.81	0.81	0.81	0.86	0.86	0.86	0.86	0.86	0.91	0.91	0.91	0.91	0.91
-4°F (-20.0°C)	TC	19722	19749	19770	19771	19797	20911	20935	20956	20960	20985	22160	22189	22213	22217	22242	23495	23521	23546	23550	23579
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.79
	PI	0.79	0.79	0.79	0.79	0.79	0.84	0.84	0.84	0.84	0.84	0.89	0.89	0.89	0.89	0.89	0.94	0.94	0.94	0.94	0.94
0°F (-17.8°C)	TC	20348	20375	20397	20398	20426	21574	21600	21621	21626	21651	22863	22893	22919	22923	22948	24241	24267	24294	24297	24328
	S/T	0.63	0.73	0.84	0.84	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	0.80	0.80	0.80	0.80	0.80	0.85	0.85	0.85	0.85	0.85	0.90	0.90	0.90	0.90	0.90	0.95	0.96	0.96	0.96	0.96
5°F (-15.0°C)	TC	21131	21159	21182	21183	21211	22404	22431	22453	22457	22484	23743	23774	23800	23804	23831	25173	25201	25228	25232	25263
	S/T	0.63	0.74	0.84	0.84	0.96	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81
	PI	0.82	0.82	0.82	0.82	0.82	0.87	0.87	0.87	0.87	0.87	0.92	0.92	0.92	0.92	0.92	0.97	0.97	0.97	0.97	0.98
14°F (-10.0°C)	TC	20732	20760	20782	20783	20811	21982	22008	22029	22034	22060	23295	23325	23351	23356	23382	24699	24725	24752	24756	24787
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79
	PI	0.87	0.87	0.87	0.87	0.87	0.92	0.92	0.92	0.92	0.92	0.98	0.98	0.98	0.98	0.98	1.04	1.04	1.04	1.04	1.04
17°F (-8.3°C)	TC	20599	20627	20649	20650	20678	21841	21867	21888	21893	21919	23146	23176	23202	23206	23232	24540	24567	24594	24598	24628
	S/T	0.62	0.72	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.88	0.55	0.64	0.73	0.73	0.83	0.52	0.61	0.69	0.69	0.79
	PI	0.89	0.89	0.89	0.89	0.89	0.94	0.94	0.94	0.94	0.94	1.00	1.00	1.00	1.00	1.00	1.06	1.06	1.06	1.06	1.06
23°F (-5.0°C)	TC	21016	21045	21067	21068	21097	22283	22309	22331	22336	22362	23614	23645	23671	23676	23702	25037	25064	25092	25095	25127
	S/T	0.61	0.72	0.81	0.81	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.90	0.90	0.90	0.90	0.90	0.95	0.95	0.95	0.95	0.96	1.01	1.01	1.01	1.01	1.01	1.07	1.07	1.07	1.07	1.07
35°F (1.7°C)	TC	21851	21880	21903	21904	21934	23167	23195	23218	23222	23250	24551	24583	24611	24615	24643	26031	26059	26087	26091	26124
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	0.92	0.92	0.92	0.92	0.92	0.98	0.98	0.98	0.98	0.98	1.03	1.04	1.04	1.04	1.04	1.10	1.10	1.10	1.10	1.10
41°F (5.0°C)	TC	21902	21931	21955	21956	21986	23222	23249	23272	23277	23305	24609	24641	24669	24673	24701	26092	26120	26149	26153	26186
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.69	0.69	0.78
	PI	0.97	0.98	0.98	0.98	0.98	1.03	1.03	1.03	1.04	1.04	1.09	1.10	1.10	1.10	1.10	1.16	1.16	1.16	1.16	1.16
47°F (8.3°C)	TC	21954	21983	22007	22008	22037	23277	23304	23327	23332	23359	24667	24699	24727	24731	24759	26153	26182	26210	26214	26247
	S/T	0.63	0.73	0.83	0.83	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	1.03	1.03	1.03	1.03	1.03	1.09	1.09	1.09	1.09	1.09	1.15	1.16	1.16	1.16	1.16	1.22	1.23	1.23	1.23	1.23
50°F (10.0°C)	TC	21919	21949	21972	21973	22003	23240	23268	23291	23295	23323	24629	24661	24688	24693	24720	26112	26141	26169	26173	26206
	S/T	0.62	0.73	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.89	0.55	0.65	0.73	0.73	0.84	0.52	0.61	0.69	0.69	0.79
	PI	1.05	1.05	1.05	1.05	1.06	1.12	1.12	1.12	1.12	1.12	1.18	1.18	1.19	1.19	1.19	1.25	1.25	1.26	1.26	1.26
59°F (15.0°C)	TC	21816	21845	21869	21870	21899	23131	23158	23181	23186	23213	24513	24545	24572	24576	24604	25990	26018	26046	26050	26083
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	1.13	1.13	1.13	1.13	1.13	1.19	1.19	1.20	1.20	1.20	1.26	1.27	1.27	1.27	1.27	1.34	1.34	1.34	1.34	1.35
68°F (20.0°C)	TC	21713	21742	21765	21766	21796	23021	23049	23071	23076	23103	24397	24428	24456	24460	24488	25867	25895	25923	25927	25959
	S/T	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78	0.48	0.57	0.65	0.65	0.74
	PI	1.20	1.20	1.20	1.20	1.20	1.27	1.27	1.27	1.27	1.28	1.35	1.35	1.35	1.35	1.35	1.43	1.43	1.43	1.43	1.43
77°F (25.0°C)	TC	21610	21639	21662	21663	21692	22912	22939	22962	22966	22994	24281	24312	24340	24344	24371	25744	25772	25800	25804	25836
	S/T	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.80	0.49	0.58	0.66	0.66	0.75	0.47	0.55	0.62	0.62	0.71
	PI	1.27	1.27	1.27	1.27	1.28	1.35	1.35	1.35	1.35	1.35	1.43	1.43	1.43	1.43	1.43	1.51	1.52	1.52	1.52	1.52
82°F (27.8°C)	TC	21886	21915	21939	21940	21970	23205	23232	23255	23260	23287	24591	24623	24651	24655	24683	26073	26101	26130	26134	26166
	S/T	0.56	0.66	0.75	0.75	0.85	0.53	0.62	0.70	0.70	0.80	0.50	0.58	0.66	0.66	0.76	0.47	0.55	0.63	0.63	0.71
	PI	1.42	1.43	1.43	1.43	1.43	1.51	1.51	1.51	1.51	1.52	1.60	1.60	1.60	1.60	1.61	1.70	1.70	1.70	1.70	1.70
86°F (30.0°C)	TC	22528	22558	22582	22583	22614	23885	23914	23937	23942	23970	25312	25345	25373	25378	25407	26837	26867	26896	26900	26934
	S/T	0.58	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79	0.49	0.58	0.65	0.65	0.75
	PI	1.69	1.69	1.69	1.69	1.69	1.79	1.79	1.79	1.79	1.80	1.90	1.90	1.90	1.90	1.90	2.01	2.01	2.02	2.02	2.02
95°F (35.0°C)	TC	23972	24004	24030	24031	24063	25416	25447	25472	25477	25507	26935	26970	27000	27005	27035	28558	28589	28620	28624	28660
	S/T	0.64	0.75	0.85	0.85	0.97	0.60	0.71	0.81	0.81											

Climate 5000 Series - Gen 4

24K Btu Single Zone Regular System - 4-Way Cassette IDU



BOSCH

Heating Performance - Indoor Dry Bulb Temperature at 765 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	/	/	/	/	/	/	/	/	/	/
-13°F(-25.0°C)	10666	10452	10400	10358	10265	1.87	1.89	1.89	1.90	1.90
-4°F(-20.0°C)	13127	12864	12800	12749	12634	1.99	2.00	2.01	2.01	2.02
0°F(-17.8°C)	15725	15410	15333	15272	15135	2.14	2.16	2.16	2.17	2.18
5°F(-15.0°C)	18972	18593	18500	18426	18260	2.33	2.35	2.36	2.36	2.37
14°F(-10.0°C)	20511	20101	20000	19920	19741	2.33	2.35	2.36	2.36	2.37
17°F(-8.3°C)	21023	20603	20500	20418	20234	2.33	2.35	2.36	2.36	2.37
22°F(-5.6°C)	20875	20457	20355	20273	20091	2.31	2.33	2.33	2.34	2.35
27°F(-2.8°C)	20726	20311	20210	20129	19948	2.29	2.31	2.31	2.32	2.33
32°F(0.0°C)	20577	20165	20064	19984	19804	2.27	2.29	2.29	2.30	2.31
35°F(1.7°C)	20487	20078	19977	19897	19718	2.26	2.27	2.28	2.28	2.29
37°F(2.8°C)	22747	22293	22181	22092	21893	2.35	2.37	2.37	2.37	2.38
42°F(5.6°C)	28398	27830	27691	27580	27332	2.57	2.59	2.60	2.61	2.62
44.6°F(7.0°C)	30658	30045	29894	29775	29507	2.67	2.69	2.69	2.70	2.71
47°F(8.3°C)	34048	33367	33200	33067	32770	2.80	2.82	2.83	2.84	2.85
52°F(11.1°C)	33574	32902	32738	32607	32313	2.60	2.63	2.63	2.64	2.65
57°F(13.9°C)	33100	32438	32276	32147	31857	2.41	2.43	2.43	2.44	2.45
59°F(15.0°C)	32910	32252	32091	31962	31675	2.33	2.35	2.35	2.36	2.37
62°F(16.7°C)	32626	31973	31814	31686	31401	2.21	2.23	2.23	2.24	2.25
64.4°F(18.0°C)	32398	31750	31592	31465	31182	2.12	2.13	2.14	2.14	2.15
69.8°F(21.0°C)	31887	31249	31093	30968	30689	1.90	1.92	1.92	1.93	1.93
75.2°F(24.0°C)	31375	30747	30593	30471	30197	1.69	1.70	1.71	1.71	1.72

Bolded values were tested. All other values are simulated and for reference only.